Jacques Reddinger

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Summary

PhD researcher specializing in Electron Microscopy and Condensed Matter Physics with expertise in magnetic-enabled electron microscopy and micromagnetic simulation paired with custom data analysis tools. Published in peer-reviewed journals, presented at national and international conferences, and recognized with awards for research and teaching excellence.

Work Experience

SCGSR Fellow Jan 2026 - present

Molecular Foundry, Lawrence Berkeley National Laboratory, Berkeley, CA

Led an independent research project on chiral spin textures in Fe/Gd multilayers, focusing on stabilization and control of antiskyrmions via interfacial Dzyaloshinskii–Moriya interaction introduced by Ir partial coverage. Performed in situ molecular beam epitaxy (MBE) of Fe, Gd, and Ir to grow hetero structures. Performed spin-polarized low-energy electron microscopy (SPLEEM) to characterize growth morphology, band structure, and surface magnetization of fabricated samples. Collaborated with beamline scientists and external researchers to integrate experimental and computational approaches; results contributing to ongoing manuscript preparation.

Graduate Research Assistant Sept 2020 - present

McMorran Lab, Dept. of Physics, University of Oregon, Eugene, OR

Conducted experimental and computational studies of the three-dimensional structure, energetics, and transport properties of topological magnetic textures in Fe/Gd multilayers. Designed and carried out experiments using advanced electron microscopy techniques including Lorentz transmission electron microscopy (LTEM) and scanning electron microscopy with polarization analysis (SEMPA). Reinforced experimental results with computational micromagnetic simulation techniques. Developed novel software for a domain tracking and energetics harvesting micromagnetics stack. Collaborated with labs at University of California San Diego's Center for Memory and Recording Research and University of Washington. Co-authored 2 peer review papers and presented at 5 international conferences. Performed regular physics outreach.

Undergraduate Research Assistant May 2019 - Aug 2020

Koeth Lab, Dept. of Physics, University of Maryland, College Park, MD

Designed and conducted an experiment studying decay via electron capture of Be-7 ions. Restored the Small Isochronous Ring (SIR), an ion storage ring, to a functioning condition. Designed and implemented various critical support systems including power control, vacuum, cooling, and structural support.

Undergraduate Research Assistant May 2019 - Aug 2020

Belloni Lab, Dept. of Physics, University of Maryland, College Park, MD

Assisted in the search for the $W\gamma$ decay of a charged Higgs boson by analyzing high-energy physics data using ROOT, Python, and C++.

Projects

Domain Tracker Link to github

Custom python package for processing the output of micromagnetic simulation software. This tool harvests perpendicular magnetic domains and their properties from standard `.ovf` files. Domains are chronologically linked into `DomainChron` objects which store morphological, topological, and energetic properties. The information harvested and structured by this tool greatly assists in the analysis of simulation results.

Education

2020 - present	PhD Physics at **University of Oregon**	(GPA: 3.96/4.0)
2016 - 2020	BS Physics at **University of Maryland**	(GPA: 3.84/4.0)

Publications

Parker, W. S., J. A. Reddinger, R. Moraski, et al. (Sept. 2025). "Real space imaging of dipole-stabilized hybrid skyrmions in magnetic multilayer thin films". In: *Physical Review B* 112.9, p. 094415.

Parker, W. S., J. A. Reddinger, et al. (Dec. 2024). "Hybrid skyrmions in magnetic multilayer thin films are half-integer hopfions". In: *Physical Review B* 110.22, p. 224420.

Conference Presentations & Posters

70th Annual Conference on Magnetism and Magnetic Materials, Palm Beach, FL, 31 October, 2025: "Soliton Formation and Topological Transitions during Stripe Collapse in Fe/Gd Multilayers"

Microscopy and Microanalysis 2025, Salt Lake City, UT, 30 July, 2025: "Dual-Modal Depth Sensitive Real Space Magnetic Imaging of 3D Skyrmions in Multilayer Thin Films"

16th Joint Conference on Magnetism and Magnetic Materials and Intermag, New Orleans, LA, 15 January, 2025: "Formation Process of Antiskyrmions in Fe/Gd Multilayers"

68th Annual Conference on Magnetism and Magnetic Materials, Dallas, TX, 1 November, 2023: "3D Dipole Antiskyrmions in Fe/Gd Multilayers Consist of Bloch Lines without Singularities"

Microscopy and Microanalysis 2022, Portland, OR, 2 August, 2022: "Micromagnetics Simulation as a Supplement to and Diagnostic for Lorentz Transmission Electron Microscopy"

Awards

2025	Science Graduate Student Research Program Awardee, Department of Energy
2025	Microscopy and Microanalysis Student Scholar Award, Microanalysis Society
2022	Best Student Poster Award, Microscopy Society of America
2022	Weiser Physics Teaching Assistant Award, University of Oregon Dept. of Physics
2016 - 2020	President's Scholarship, University of Maryland

Technical Proficiencies

Electron Microscopy	SEM, SEMPA, TEM, LTEM, STEM, SPLEEM
Spectroscopy/Crystallography	Auger spectroscopy, LEEM I-V, LEED, EDS, XRD, XRR
Nanofabrication	FIB, MBE, sputter deposition, plasma cleaning, ion milling
Experimental Apparatus	UHV, electron/ion optics and detectors, power and measurement electronics
Programming	Python, Javascript, machine learning, image analysis, computer vision
Simulation	Micromagnetics, custom Monte-Carlo simulations